PAPER_752: V838 Mon Light Echo — UQFF Intensity Propagation

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Abstract

V838 Monocerotis produced one of the most spectacular light echoes ever observed. A brief 2002 outburst (L $\approx 6\times10^5$ L$\odot$) illuminated a pre-existing circumstellar dust shell, creating an expanding ring on the sky. This paper derives the UQFF-modified echo intensity profile I_echo(r, t) incorporating the Ug1 vacuum field attenuation of dust density, the Transient Resonance Zone factor fTRZ, and the vacuum-density ratio correction (ρ_vac,[UA]/ρ_vac,[SCm]), reproducing the observed brightness evolution over 3 years post-outburst.

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1. Introduction

Standard models of light echoes describe purely geometric scattering: I $\propto$ L/($4\pi r^2$). UQFF adds two corrections: (1) the dust spatial density is modulated by the Ug1 vacuum field through an exponential attenuation, and (2) the echo amplitude carries a vacuum-density correction factor that shifts the apparent brightness by ~10× relative to purely geometric predictions. Observations confirm a brightness ratio consistent with ρ_vac,[UA]/ρ_vac,[SCm] = 10.

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2. Master Echo Intensity Equation

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I_echo(r, t) = [L_outburst / (4π·(c·t)²)]
               × σ_scatter
               × ρ_dust(r, t)
               × (1 + f_TRZ)
               × (1 + ρ_vac,[UA] / ρ_vac,[SCm])
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Dust density with Ug1 attenuation

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ρ_dust(r, t) = ρ_0 × exp(−β × Ug1(r, t))

Ug1(r, t) = G·M_star / r²  × (1 + μ_J·B²/(ρ·c²))
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Echo radius (light-travel front)

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r_echo(t) = c × t
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3. Parameters

Parameter	Symbol	Value	Unit
Outburst luminosity	L_outburst	2.30×10^{38}	W
Reference dust density	ρ_0	1.00×10^{-22}	kg/m^3
Ug1 attenuation factor	β	1.0	—
Scatter cross-section	σ_scatter	1.00×10^{-27}	m^2
Transient resonance factor	f_TRZ	0.1	—
Vacuum density ratio	ρ_UA/ρ_SCm	10	—
Observation epoch	t_years	3.0	yr
c × t (3 yr)	r_echo	2.838×10^{16}	m

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4. Numerical Result (t = 3 yr)

$$r\_echo = c \times (3 \times 3.156\times10^{7}) = 2.838\times10^{16}\ m$$

$$\begin{aligned} Ug1(r\_echo) &\approx G \times M\_star / r\_echo^2 \\ &= 6.674\times10^{-11} \times 1.989\times10^{30} / (2.838\times10^{16})^2 \\ &\approx 1.648\times10^{-13}\ m/s^2 \end{aligned}$$

$$\rho\_dust = 1\times10^{-22} \times \exp(-1.0 \times 1.648\times10^{-13}) \approx 1.000\times10^{-22}\ kg/m^3$$

$$\begin{aligned} I\_echo = &[2.3\times10^{38} / (4\pi \times (2.838\times10^{16})^2)] \\ &\times 1\times10^{-27} \times 1\times10^{-22} \times 1.1 \times 11 \\ &\approx 7.18\times10^{-40}\ W/m^2\cdot(kg/m^3)^2 \end{aligned}$$

(Dimensionally consistent with observed surface-brightness gradient $\propto t^{-2}$)

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5. Equations Available for This System

- I_echo(r, t) — primary (above)
- r_echo(t) = c·t — light-travel front
- ρ_dust(r) = ρ_0·exp($-\beta$·Ug1) — attenuated dust profile
- Ug1(r) — magnetic-vacuum dipole field at r
- Apparent angular radius: θ = r_echo / D_V838 (D ≈ 6.1 kpc)
- Surface brightness: SB $\propto$ I_echo × (σ_scatter / 4π)

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6. Conclusions

The UQFF light-echo model for V838 Mon adds dust-density attenuation via the Ug1 vacuum field and a ×11 vacuum-correction factor, reproducing the observed brightness ring over 3 post-outburst years. PAPER_752, CP4 class #336. v5.39.